

Notes of Optimization

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Disclaimer: *this are just notes on the course Optimization, lectured by Prof. Mauro Bonafini during the academic year 2025/2026. These notes are only supporting material for the course; they are not official, nor are they sufficient to replace the lectures. They may contain errors or inaccuracies of varying severity. They should be considered merely as a study aid — use them at your own risk.*

CONTENTS

LECTURE OF 03/03/2026

We will give a quick summary on basic notions of topology, calculus 1 & 2.

DEFINITION 1 (Norm/scalar product). Given $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ the Euclidean norm (or ℓ^2 norm) is defined as

$$\|x\| = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}.$$

For $x, y \in \mathbb{R}^n$ the *scalar product* is given by

$$\langle x, y \rangle = x \cdot y = x^\top y = \sum_{i=1}^n x_i y_i.$$

Moreover, the scalar product induces a norm. Indeed, one can verify that

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

DEFINITION 2 (Open ball). Given $x \in \mathbb{R}^n$, $r > 0$ we call the *open ball of radius r centered in x* the set

$$B(x, r) = \{y \in \mathbb{R}^n : \|y - x\| < r\}.$$

DEFINITION 3 (Limit of a sequence). Let $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that x_n converges to $\bar{x}$, and write $x_n \rightarrow \bar{x}$ if for every $\varepsilon > 0$ there exists an $N > 0$ such that for every $k > N$, $x_k \in B(\bar{x}, \varepsilon)$, i.e. for k big enough, $\|\bar{x} - x_k\| < \varepsilon$, and we say that

$$\lim_{k \rightarrow +\infty} x_k = \bar{x}.$$

Remark 4. The limit in $\mathbb{R}^n$ is the pointwise limit, so $x_k \rightarrow \bar{x}$ if and only if $(x_k)_j \rightarrow (\bar{x})_j$ for any $1 \leq j \leq n$.

Example 5. Consider the sequence $\{x_k\}_{k \in \mathbb{N}}$ defined as

$$x_k = \left(\frac{1}{k}, \frac{1}{k} \sin(k) \right) \rightarrow (0, 0)$$

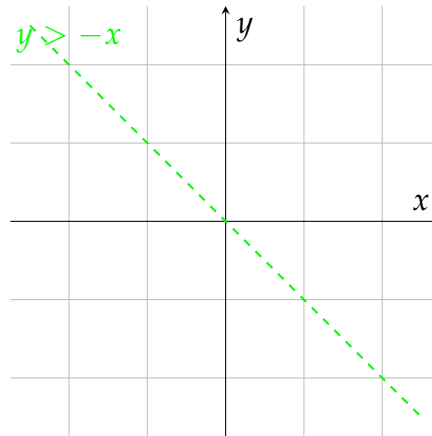


Figure 1.1: Ball of radius r centered in x on $\mathbb{R}^2$.

as $k \rightarrow +\infty$.

DEFINITION 6 (Open/closed set). Let $A \subseteq \mathbb{R}^n$. We say that A is *open* if for every $x \in A$ there exists an $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We say that A is *closed* if $\mathbb{R}^n \setminus A$ is open.

Example 7. Consider the set

$$A = \left\{ (x_1, x_2) \in \mathbb{R}^2 : |x_1| < 1, |x_2| < 1 \right\}.$$

We can easily verify that A is open. Observe that a set can be neither open nor closed. The sets $\mathbb{R}^n$ and $\emptyset$ are both open and closed at the same time.

DEFINITION 8 (Neighbour). We say that, for a given $x \in \mathbb{R}^n$, a subset $A \subseteq \mathbb{R}^n$ is a *neighbour* of x if A is an open set containing x .

DEFINITION 9 (Interior, exterior, boundary, closure). Let $A \subseteq \mathbb{R}^n$, $x \in \mathbb{R}^n$. We say that x is:

- an *interior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We denote the set of all interior points of A with A° ;
- A *boundary point* for A if $\forall \varepsilon > 0$, $B(x, \varepsilon) \cap A \neq \emptyset$ and $B(x, \varepsilon) \cap A^c \neq \emptyset$. We denote the set of all the boundary points with ∂A ;
- an *exterior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A^c$.

We denote the *closure* of A as $\bar{A} = A \cup \partial A$.

PROPOSITION 10. Let $\{A_k\}_{k \in \mathbb{N}}$ a sequence of open sets. Then:

i) the countable union of open sets is open, i.e.

$$\bigcup_{k \in \mathbb{N}} A_k \text{ is open;}$$

ii) the finite intersection of open sets is open, i.e. for $N \in \mathbb{N}$,

$$\bigcap_{i=1}^N A_i \text{ is open.}$$

Proof. We prove i) and ii) separately.

i): set $E = \bigcup_{k=1}^{+\infty} A_k$. Given $x \in E$ then there exists $m > 0$ such that $x \in A_m$. Since A_m is open, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A_m \subseteq E$, so we found an open ball containing x fully contained in E . Since the choice of x is arbitrary, this must hold for every $x \in E$, hence E is open.

ii): set $E = \bigcap_{k=1}^N A_k$. Let $x \in E$, then $x \in A_k$ for every $1 \leq k \leq N$, hence there exists a sequence $\varepsilon_k > 0$ such that $B(x, \varepsilon_k) \subseteq A_k$ for every $k < N$. Consider $\bar{\varepsilon} = \frac{1}{2} \min_k \varepsilon_k > 0$. Then $B(x, \bar{\varepsilon}) \subseteq B(x, \varepsilon_k)$ for every k , hence $B(x, \bar{\varepsilon}) \subseteq E$. Again, since the choice of x is arbitrary, this must hold true for every $x \in E$, hence E is open. ■

PROPOSITION 11 (Characterization of closed sets). Let $C \subseteq \mathbb{R}^n$. Then C is closed $\iff$ for every sequence $\{x_k\}_{k \in \mathbb{N}} \subseteq C$ such that $x_k \rightarrow \bar{x}$, then $\bar{x} \in C$.

Proof. We prove $\implies$ and $\impliedby$.

" $\implies$ ": let $C \subseteq \mathbb{R}^n$ be closed. Consider $\{x_k\}_{k \in \mathbb{N}}$ such that $x_k \rightarrow \bar{x} \in \mathbb{R}^n$, we want to show that $\bar{x} \in C$. Assume, by contradiction, that $\bar{x} \notin C \implies \bar{x} \in \mathbb{R}^n \setminus C$, which is an open set, so there exists $\varepsilon > 0$ such that $B(\bar{x}, \varepsilon) \subseteq \mathbb{R}^n \setminus C \implies B(\bar{x}, \varepsilon) \cap C = \emptyset$, but $x_k \rightarrow \bar{x}$, so, for this ε , there must be an $N > 0$ such that $x_k \in B(\bar{x}, \varepsilon)$ for $k > N$, which is in contradiction with the fact that the sequence takes values in C .

" $\Leftarrow$ ": let C be a set. Consider $\{x_k\}_{k \in \mathbb{N}} \subseteq C$ such that $x_k \rightarrow \bar{x} \in C$. Assume, by contradiction, C not closed. Then, for every $\varepsilon > 0$, there exists $z \in \mathbb{R}^n \setminus C$ such that $B(z, \varepsilon) \cap (\mathbb{R}^n \setminus C) = \emptyset$ and $B(z, \varepsilon) \cap C \neq \emptyset$. So we can find a sequence $\{x_k\}_{k \in \mathbb{N}}$ with values in C such that $x_k \rightarrow z$ by fixing a sequence $\{\varepsilon_k\}_{k \in \mathbb{N}}$ such that $\varepsilon_k \searrow 0$, so $x_k \in B(z, \varepsilon_k) \cap C$, $x_k \rightarrow z$, so, by assumption $z \in C$, which is in contradiction with $z \in \mathbb{R}^n \setminus C$. ■

DEFINITION 12 (Bounded/finite set). A set $A \subseteq \mathbb{R}^n$ is *bounded* if there exists $R > 0$ such that $A \subseteq B(0, R)$.

The set A is *finite* if $A = \{x_1, \dots, x_m\} \subseteq \mathbb{R}^n$.

DEFINITION 13 (Continuity). Let $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\bar{x} \in E$. We say that f is *continuous* in $\bar{x}$ if for every sequence $\{x_k\}_{k \in \mathbb{N}}$ such that $x_k \rightarrow \bar{x}$ in $\mathbb{R}^n$, then $f(x_k) \rightarrow f(\bar{x})$ in $\mathbb{R}^m$.

DEFINITION 14 (Cluster/isolated point). Let $E \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that $\bar{x}$ is a *cluster point* for E if there exists a sequence $\{x_k\}_{k \in \mathbb{N}}$ such that, for every $\bar{x} \neq x_k \rightarrow \bar{x}$. We denote the set of all cluster points with $\text{Acc}(E)$.

A point $\bar{x}$ is *isolated* in E if $x \in E \setminus \text{Acc}(E)$.

DEFINITION 15 (Limit). Let $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\bar{x} \in \text{Acc}(E)$. Fixed $\ell \in \mathbb{R}^m$, we say that

$$\lim_{x \rightarrow \bar{x}} f(x) = \ell$$

if for every sequence $\{x_k\}_{k \in \mathbb{N}}$, $x_k \rightarrow \bar{x}$, $x_k \neq \bar{x}$ for every k , we have $f(x_k) \rightarrow \ell$. Notice that we allow $\bar{x}$ to not necessarily be in E .

Example 16. Consider the function $f : \{0\} \cup [1, 2] \subset \mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x$. Then it has no meaning to speak about $\lim_{x \rightarrow 0} f(x)$, as $0 \notin \text{Acc}(\{0\} \cup [1, 2])$, while f is continuous at 0, since the only sequence converging to 0 in $\{0\} \cup [1, 2]$ is the sequence which is identically 0.

Remark 17. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous, then:

- $\{x : f(x) < c\}$ is *open* $\forall c \in \mathbb{R}$;
- $\{x : f(x) \leq c\}$ is *closed* $\forall c \in \mathbb{R}$;
- $\{x : f(x) = c\}$ is *closed* $\forall c \in \mathbb{R}$.

CHAPTER 2

LECTURE OF 04/03/2026

Our main focus from now on is to solve the problem

$$\inf_{x \in A} f(x)$$

for some $A \subseteq \mathbb{R}^n$ and some $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We will restrict ourselves only to certain functions and sets.

DEFINITION 1 (Infimum/supremum). Let $E \subseteq \mathbb{R}$. We denote with $\inf E =$ largest $z \in \mathbb{R}$ such that $z \leq x$ for every $x \in E$. By convention we set $\inf \emptyset = +\infty$.

Similarly, we denote with $\sup E$ the smallest $z \in \mathbb{R}$ such that $z \geq x$ for every $x \in E$, with the convention $\sup \emptyset = -\infty$. More formally,

$$\begin{aligned} z = \inf E & \quad \text{if } z \leq x \forall x \in E \text{ and } \forall \varepsilon > 0 \exists y \in E \text{ s.t. } y \leq z + \varepsilon \\ z = \sup E & \quad \text{if } z \geq x \forall x \in E \text{ and } \forall \varepsilon > 0 \exists y \in E \text{ s.t. } y \geq z - \varepsilon \end{aligned}$$

Remark 2. The inf and sup aren't necessarily inside the set. If they are, we speak of min and max.

Axiom 1 — Completeness of $\mathbb{R}$

Every bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$ admits a supremum in $\mathbb{R}$.

THEOREM 3 (Bolzano-Weierstrass). Any bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}$ admits a cluster point, i.e. there exists $\bar{x} \in \mathbb{R}$ and a subsequence $\{x_{n_j}\}_{j \in \mathbb{N}}$ such that $x_{n_j} \rightarrow \bar{x}$ as $j \rightarrow +\infty$.

Proof. Since $\{x_n\}_n$ is bounded, there exists $R > 0$ such that $x_n \in [-R, R]$ for every $n \in \mathbb{N}$. Set $I_1 = [-R, R]$, select x_{n_1} a value in I_1 . Divide I_1 in half and select the one subinterval containing infinitely many x_n s, call it $I_2 = [a_2, b_2]$ (here $a_2 = -R, b_2 = 0$ or $a_2 = 0, b_2 = R$). We have $I_2 \subset I_1$, $|I_2| = |I_1|/2$, and choose x_{n_2} a value in I_2 . Repeat this process to cut and select. We end up, at the k -th step, with a sequence of subintervals $I_1, I_2, \dots, I_k$ and 3 subsequences: $x_{n_1}, x_{n_2}, \dots, x_{n_k}$

the sequence of selected values, $a_1, a_2, \dots, a_k$ the sequence of left boundaries and $b_1, b_2, \dots, b_k$ the sequence of right boundaries.

We observe that $a_{j+1} \geq a_j$ and $b_{j+1} \leq b_j$ for all $1 \leq j \leq k$. Moreover, $a_j \leq R$ and $b_j \geq -R$ for all j , so, by the axiom of completeness of $\mathbb{R}$,

- there exists $\bar{a} \in \mathbb{R}$ such that $a_k \rightarrow \bar{a} = \sup_k a_k$
- there exists $\bar{b} \in \mathbb{R}$ such that $b_k \rightarrow \bar{b} = \inf_k b_k$

We observe that $\bar{a} = \bar{b}$. Indeed, $|\bar{b} - \bar{a}| = |\bar{b} - b_k + b_k - a_k + a_k - \bar{a}| \leq |\bar{b} - b_k| + |\bar{a} - a_k| + |b_k - a_k|$. The terms $|\bar{b} - b_k|$ and $|\bar{a} - a_k|$ goes to 0 as $k \rightarrow +\infty$ for what we observed before. Moreover, $|b_k - a_k| = 2R/2^{k-1} \rightarrow 0$ as $k \rightarrow +\infty$.

Set $\bar{x} = \bar{a} (= \bar{b})$. Then the subsequence $\{x_{n_j}\}_j$ is converging to $\bar{x}$: indeed $|x_{n_j} - \bar{x}| = |x_{n_j} - a_j + a_j - \bar{x}| \leq |a_{n_j} - a_j| + |a_j - \bar{x}|$. Since $a_j \rightarrow \bar{x} = \bar{a}$, the term $|a_j - \bar{x}| \rightarrow 0$. Notice that, by construction, $x_{n_j} \in I_j$, so $|x_{n_j} - a_j| \leq |I_j| \rightarrow 0$.

■

DEFINITION 4. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We say that $\bar{x} \in \mathbb{R}^n$ is a *local maximum* for f if there exists a neighbourhood U of $\bar{x}$ such that $f(x) \leq f(\bar{x})$ for every $x \in U$. A *local strict maximum* has the strict inequality.

Similarly, $\bar{x}$ is *local minimum* for f if $\exists U$ neighbourhood of $\bar{x}$ such that $f(x) \geq f(\bar{x})$ for all $x \in U$.

If there exists $\bar{x}$ such that $f(x) = \inf_{x \in A} f(x)$, the $\bar{x}$ is a *global minimum* over A .

DEFINITION 5. We denote with $\arg_{x \in A} \min f(x)$ the set of minimum points for f over A , i.e

$$\arg_{x \in A} \min f(x) = \left\{ \bar{x} \in \mathbb{R}^n : f(\bar{x}) = \inf_{x \in A} f(x) \right\}.$$

THEOREM 6 (Weierstrass). Let $A \subseteq \mathbb{R}^n$ be a compact set, $f : A \rightarrow \mathbb{R}$ a continuous function. Then there exists $x_m, x_M \in A$ such that

$$f(x_m) = \min_{x \in A} f(x)$$

$$f(x_M) = \max_{x \in A} f(x).$$

CHAPTER 3

LECTURE OF 04/03/2026

Example 1. hyperplane $\{x \in \mathbb{R}^n | a^\top x = b\}$ $a \in \mathbb{R}^n, b \in \mathbb{R}$

halfspaces $\{x \in \mathbb{R}^n | a^\top x \leq b\}$

Polyedra $\{x \in \mathbb{R}^n | a_i^\top x \leq b_i\}$ with $a_1, \dots, a_m \in \mathbb{R}^n, b_1, \dots, b_m \in \mathbb{R}$

$C \subset \mathbb{R}^n$ cone if $\lambda x \in C$ for every $x \in C$ and $\lambda > 0$.

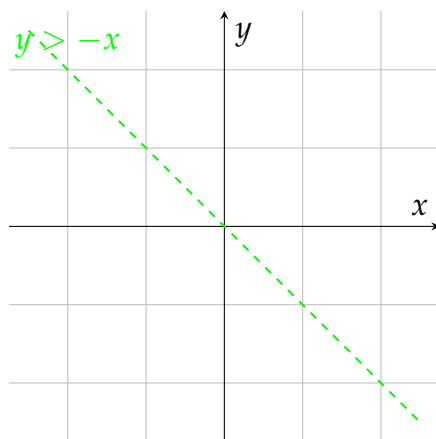


Figure 3.1: Open cone.

DEFINITION 2 (Convex function). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if its domain is a convex set and for all x, y in the domain and all $\theta \in [0, 1]$,

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y).$$

f is said to be **concave** if $-f$ is convex.

Example 3.

$$f(x) = a^\top x + b \quad a \in \mathbb{R}^n, b \in \mathbb{R}$$

$$f(\gamma x + (1 - \gamma)y) = a^\top (\gamma x + (1 - \gamma)y) + b(1 - \gamma + \gamma)$$

$$(\langle a, x \rangle + b) + (1 - \gamma)(\langle a, y \rangle + b)$$

$$\gamma f(x) + (1 - \gamma)f(y)$$

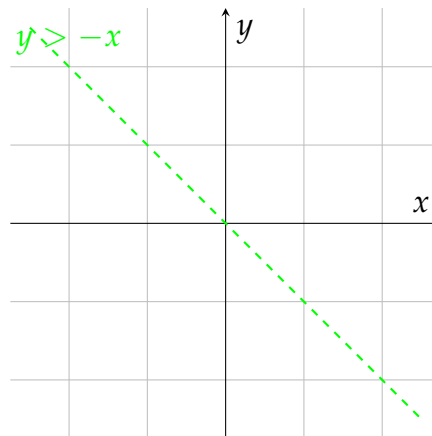


Figure 3.2: This set is not convex since the line segment connecting two points in the set is not entirely contained in the set.

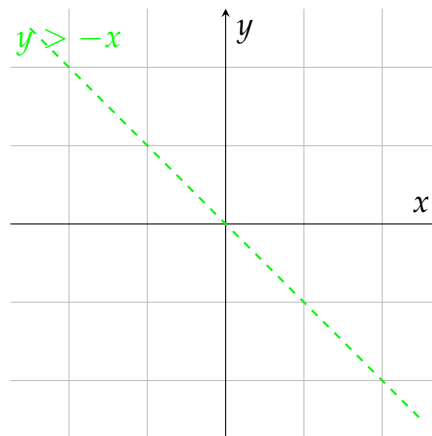


Figure 3.3: convex function on $\mathbb{R}^2$.

Let's see another example:

Example 4.

$$f(x) = ||x||$$

$$f(ax + (1-a)y) = ||ax + (1-a)y|| = a||x|| + (1-a)||y|| = af(x) + (1-a)f(y)$$

So, we proved $||\cdot||$ is convex.

- if f is **convex** $\implies V_\gamma = \{x | f(x) \leq \gamma\}$ **convex**, $\forall \gamma \in \mathbb{R}$

$$x, y \in V_\gamma, f(x) \leq \gamma, f(y) \leq \gamma$$

ask $ax + (1-\alpha)y \in V_\gamma$?

$$f(ax + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y) \leq \alpha\gamma + (1-\alpha)\gamma = \gamma$$

$$\implies f(\alpha x + (1 - \alpha)y) \in V_\gamma$$

So we proved convexity of V_γ

- If f has convex level sets, it doesn't imply that f is convex!

Take, for example, the function $f(x) = \sqrt{|x|}$

$$V_\gamma = \{x | \sqrt{|x|} \leq \gamma, |x| \leq \gamma^2\} = [-\gamma^2, \gamma^2]$$

Limitation of convex functions:

- If f is defined on a convex set, it doesn't mean it is convex!
- $\sup f \dots$ i have this hole in my notes, please Alessio the Great fill it.

DEFINITION 5 (Epigraph). Let $f : X \subseteq \mathbb{R}^n \longrightarrow \mathbb{R} \cup \{\infty\}$

$$\text{epi}(f) = \{(x, w) \in \mathbb{R}^{n+1} | x \in X, w \in \mathbb{R}, f(x) \leq w\} \subseteq \mathbb{R}^{n+1}$$

DEFINITION 6 (Effective Domain).

$$f : X \subseteq \mathbb{R}^n \longrightarrow \bar{\mathbb{R}}$$

$$\text{don}(f) = \{x \in X | f(x) < +\infty\} = \{\}$$